

Siddheya Patade

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Education

Georgia Institute of Technology Atlanta, GA	Expected May 2027
<i>Master of Science, Computational Data Analytics</i>	
Stockton University Galloway, NJ	2024
<i>Bachelor of Science, Business Management</i>	

Technical Skills

Languages: Python, Advanced SQL (window functions, CTEs, query optimization), TypeScript / JavaScript, R, VBA, Rust (familiar)
ML & AI: XGBoost, Scikit-learn, TensorFlow, PyTorch, NLP, Document Parsing, Multi-Agent / Agentic Systems, OpenAI API, Local LLM Inference (Ollama), Model Quantization, Markov Chains, Graph Network Analysis
AI-Driven Development: Claude Code, Cursor, agentic coding workflows, prompt engineering, LLM evaluation harnesses, spec-driven development, test-driven AI workflows
Data Engineering: Dagster, Airflow, ETL / ELT Pipelines, Data Modeling (curated / ADS layers, semantic layers), Data Validation, Unit Testing (pytest), Root-Cause Analysis, Audit Logging, FastAPI, Flask, Next.js, REST, WebSockets
Databases & Warehouses: BigQuery, SQL Server, PostgreSQL, Redshift, MySQL, MongoDB, SQLite, SpatiaLite, Trino; Tableau, Power BI (DAX), Looker, Matplotlib, Seaborn
Cloud & DevOps: GCP (BigQuery, Cloud Run, Workload Identity Federation), AWS (S3, Lambda, Redshift), Azure, Docker, Kubernetes, GitHub Actions, Jenkins, Git, Firebase, Gurobi (optimization)

Experience

NPPG (Pinnacle) Shrewsbury, NJ (Remote)	July 2025 – Present
<i>Data Analyst, Data Engineering & Machine Learning</i>	
- Sole technical hire who built the firm's analytics capability from zero using AI-driven development workflows (Claude Code, spec-driven prompting, agentic refactors); platform is used daily by 10 senior executives for risk triage of 13,905 financial plans managing \$500M+ in assets, replacing manual spreadsheet workflows. - Engineered 24 automated Dagster pipelines (88 assets, 24 scheduled jobs) producing curated / ADS-style datasets that feed downstream risk scoring, Markov chain modeling, graph network analysis, and workload optimization; instrumented unit tests, data validation checks, and structured logging so pipeline failures surface with actionable root-cause context. - Developed 7 production ML models (GradientBoosting, RandomForest, XGBoost) with automated 3-model shootout and 5-fold CV; risk classifier hits 0.911 AUC / 82% recall, churn model 0.986 AUC / 92% recall; system generates 53K predictions per nightly run. Out-of-time backtesting showed reviewing 12% of plans catches 39% of failures (3.3x triage lift). - Shipped full-stack triage dashboard (Next.js, FastAPI, SQL Server) with interactive graph-network views; built Power BI platform giving the CFO first-ever visibility into bottlenecks across 13K+ work units (cut board reporting prep 80%); codified regulatory logic into an automated rules engine with audit trails (each miss = up to \$500K penalty).	
Handshake (Project Helix) Remote	April 2026 – Present
<i>Software Engineer, Agentic Coding Evaluation</i>	
- Build reproducible Docker-based evaluation harnesses (helix-validation) for agentic-coding model training across major OSS codebases (bevyengine/bevy, cashapp/misk, excalidraw, graphql-java) spanning Rust, Kotlin/JVM, TypeScript, and Python toolchains. - Author golden-solution implementations and pin per-task test scopes (Gradle subprojects, Cargo packages); harden CI seeding by switching base images, baking Gradle deps into image build, and rewriting multi-module runners for monorepo path layouts.	
NPPG (Pinnacle) East Brunswick, NJ	December 2024 – July 2025
<i>Implementation Specialist</i>	
- Architected document parsing + comparison engine processing 100+ retirement plan documents in 200s (99.7% cycle-time reduction from manual 1,000+ minutes, 1,800 plans/hour); CFO estimated combined parser + compliance integration delivers ~3 FTE reclaimed and ~5% net revenue uplift. - Deployed internal audit tools with Python GUIs so non-technical users could run data validation and reporting independently.	
United States Marine Corps Various Locations	May 2020 – 2024
<i>Bulk Fuel Operations Specialist, Logistics & Operations Analytics</i>	
Managed inventory tracking and accountability for up to 1 million gallons of fuel across multiple systems; built performance tracking and corrective-action documentation processes for the unit.	

Projects

Plan Matrix	github.com/spatadenppg/googlyeyes
<i>Python, FastAPI, Next.js 16, PyMuPDF, OpenCV, Ollama (qwen2.5vl)</i>	
- Cross-document field comparison + IRS-compliance review tool for 401(k) adoption agreements; ingests AcroForm, flattened, and pure-scan PDFs through three auto-routed extraction backends (PyMuPDF, OpenCV, local Ollama vision) onto a canonical schema with per-leaf provenance and confidence. - Declarative YAML rule catalog with a deterministic mini-DSL so compliance officers sign off without reading source; every issue carries inline IRC / CFR / NPPG base-plan citations linked to the cited PDF page. Excel + WeasyPrint PDF exports with per-rule CI fixtures.	
Electric Truck Charging Point Optimization	github.com/inSideos-designs/ChargingPointOptimization
<i>Python, Dagster, SpatiaLite, Gurobi, GRAPE, Leafmap</i>	
- End-to-end pipeline recommending 100 optimal new heavy-duty truck DC fast-charging sites across the lower 48, joining 10 public datasets (FAF5 freight flows, HPMS highway geometry, AFDC stations, EIA grid, NLCD land cover, OSM, Census TIGER, BTS CFS) into SpatiaLite via a four-layer Dagster asset DAG. - Built a ~200K-edge weighted freight network in GRAPE; ran four Gurobi facility-location formulations (CFLP, P-Median, Set Cover, Flow-Refueling) under shared budget + vehicle-range constraints to produce a consensus ranking, validated against the AFDC baseline. Georgia Tech CSE 6242 capstone, Spring 2026.	